

CC 352

Computer Graphics

Lecture 9

Two dimensional geometric
transformation

(Basic Two-Dimensional Geometric
Transformations)

Use of Transformations

- Map points from one coordinate system to points in another coordinate system
- Change the shape of objects
- Position objects in a scene
- Create multiple copies of objects in the scene
- Projection for virtual cameras
- Animations

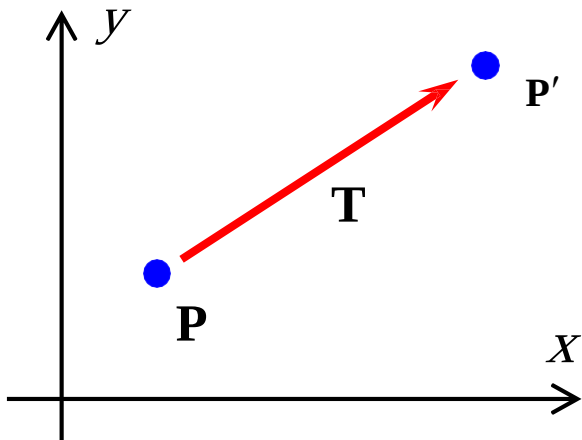


Basic Two-Dimensional Geometric Transformations

- Basic geometric transformations:
 - Translation Change position
 - Rotationchange orientation
 - Scaling change size
- Other useful transformations:
 - Reflection mirror
 - Shear change shape

(a) 2D Translation

- Translation is the straight line movement of an object from its original position to another position at a given distance and direction
- Let **P** be the input vertex and **tr** be the shifts in **x** and **y** directions, respectively. Recall the equation of transformation **P' = P + tr** . Therefore, the translated vertex is given by



$$x' = x + t_x, \quad y' = y + t_y$$

$$\mathbf{P} = \begin{bmatrix} x \\ y \end{bmatrix}, \quad \mathbf{P}' = \begin{bmatrix} x' \\ y' \end{bmatrix}, \quad \mathbf{T} = \begin{bmatrix} t_x \\ t_y \end{bmatrix}$$

$$\mathbf{P}' = \mathbf{P} + \mathbf{T}$$



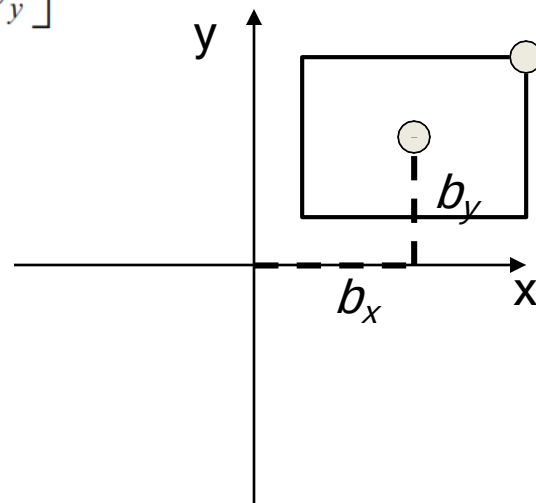
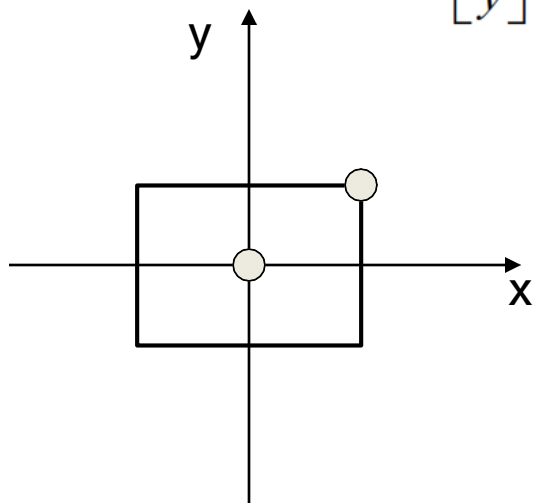
2D Translation

- Moves an object

$$x' = x + t_x, \quad y' = y + t_y$$

$$\mathbf{P} = \begin{bmatrix} x \\ y \end{bmatrix}, \quad \mathbf{P}' = \begin{bmatrix} x' \\ y' \end{bmatrix}, \quad \mathbf{T} = \begin{bmatrix} t_x \\ t_y \end{bmatrix}$$

$$\mathbf{P}' = \mathbf{P} + \mathbf{T}$$



(a) 2D Translation:

```
void translate (point [ ], int n, int tx, int ty)
{
    int i;
    for (i = 0; i < n; i++)
    {
        point[i].x = point[i].x + tx;
        point[i].y = point[i].y + ty;
    }
}
```

Example

- Given a square with coordinate points A(0, 3), B(3, 3), C(3, 0), D(0, 0). Apply the translation with distance 1 towards X axis and 1 towards Y axis. Obtain the new coordinates of the square..

For Coordinates A(0, 3)

$$X_{\text{new}} = X_{\text{old}} + T_x = 0 + 1 = 1$$

$$Y_{\text{new}} = Y_{\text{old}} + T_y = 3 + 1 = 4$$

For Coordinates B(3, 3)

$$X_{\text{new}} = X_{\text{old}} + T_x = 3 + 1 = 4$$

$$Y_{\text{new}} = Y_{\text{old}} + T_y = 3 + 1 = 4$$

For Coordinates C(3, 0)

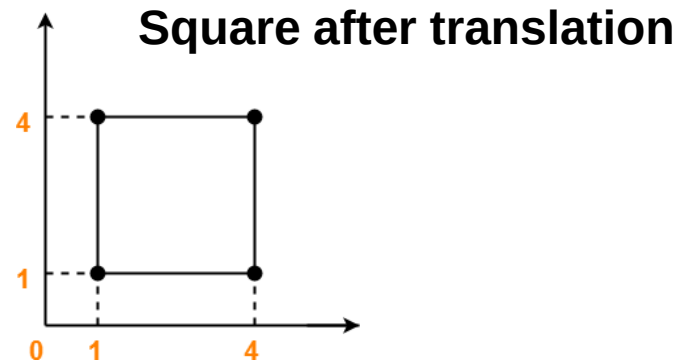
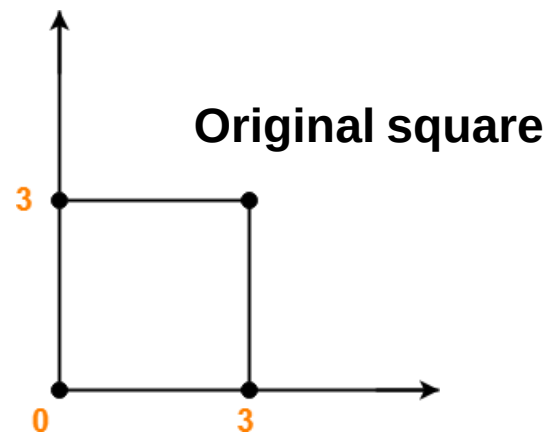
$$X_{\text{new}} = X_{\text{old}} + T_x = 3 + 1 = 4$$

$$Y_{\text{new}} = Y_{\text{old}} + T_y = 0 + 1 = 1$$

For Coordinates D(0, 0)

$$X_{\text{new}} = X_{\text{old}} + T_x = 0 + 1 = 1$$

$$Y_{\text{new}} = Y_{\text{old}} + T_y = 0 + 1 = 1$$





Cartesian and Polar Coordinates

If $\overline{PL} \perp$ x-axis with $\overline{PL} = y_1$ and $\overline{PM} \perp$ y-axis with $\overline{PM} = x_1$.

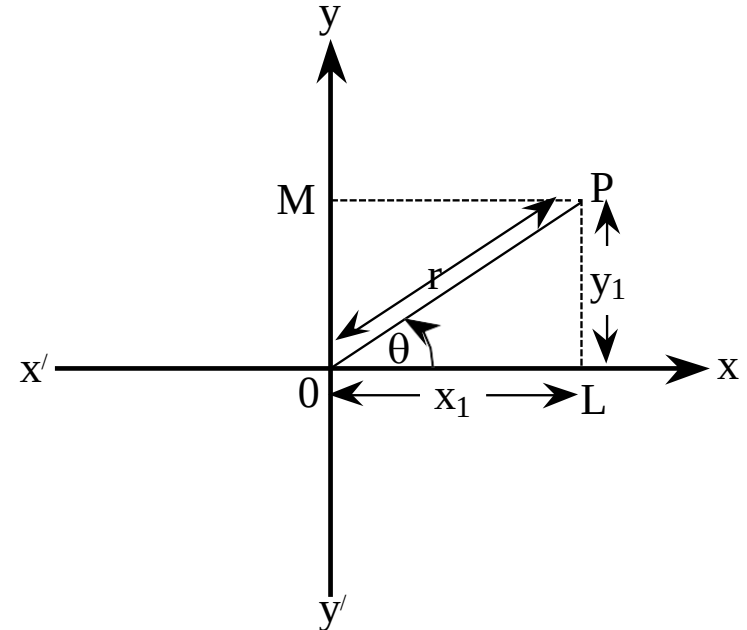
Then (x_1, y_1) are called the Cartesian coordinates of the point P.

If $OP = r$ and $\angle LOP = \theta$, then (r, θ) are called the **polar coordinates** of the point P.

It is clear that

$$x_1 = r \cos \theta, \quad y_1 = r \sin \theta,$$

$$r = \sqrt{x_1^2 + y_1^2} \quad \text{and} \quad \theta = \tan^{-1} \frac{y_1}{x_1}$$





Example 1

Find the polar coordinates of the point with Cartesian coordinates (1, 1).

Solution

$$r = \sqrt{x^2 + y^2} = \sqrt{2}, \quad \theta = \tan^{-1} \frac{y}{x} = \tan^{-1} 1 = \frac{\pi}{4},$$

i.e. the polar coordinates are: $\left(\sqrt{2}, \frac{\pi}{4} \right)$

Example 2

Find the Cartesian coordinates of the point with polar coordinates $\left(4, \frac{3\pi}{4} \right)$

Solution

$$x = r \cos \theta = 4 \cos \frac{3\pi}{4} = 4 \left(-\frac{1}{\sqrt{2}} \right) = -2\sqrt{2},$$

$$y = r \sin \theta = 4 \sin \frac{3\pi}{4} = 4 \left(\frac{1}{\sqrt{2}} \right) = 2\sqrt{2},$$

i.e. the Cartesian coordinates are: $(-2\sqrt{2}, 2\sqrt{2})$

Basic 2D Geometric Transformations

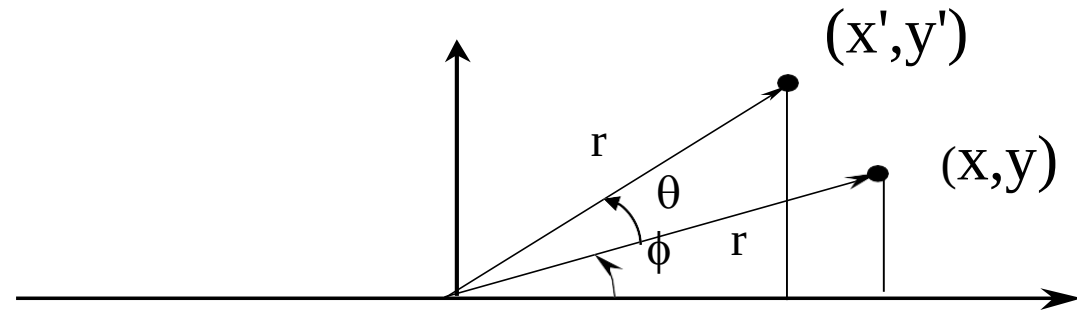
(2D Rotation)

- 2D Rotation
 - If θ is positive $\rightarrow$ counterclockwise rotation
 - If θ is negative $\rightarrow$ clockwise rotation
 - Remember:
 - $\cos(a + b) = \cos a \cos b - \sin a \sin b$
 - $\cos(a - b) = \cos a \sin b + \sin a \cos b$

(b) 2D Rotation

- Rotation is the transformation along circular paths. Rotation angle determines the amount of rotation for each vertex.

$$x = r \cos \phi$$
$$y = r \sin \phi$$



$$x' = r \cos(\phi + \theta) \quad , \quad y' = r \sin(\phi + \theta)$$

$$x' = r \cos \theta \cos \phi - r \sin \theta \sin \phi \Rightarrow x' = x \cos \theta - y \sin \theta$$

$$y' = r \sin \theta \cos \phi + r \cos \theta \sin \phi \Rightarrow y' = x \sin \theta + y \cos \theta$$

$$\mathbf{P}' = \mathbf{R} \cdot (\mathbf{P})$$

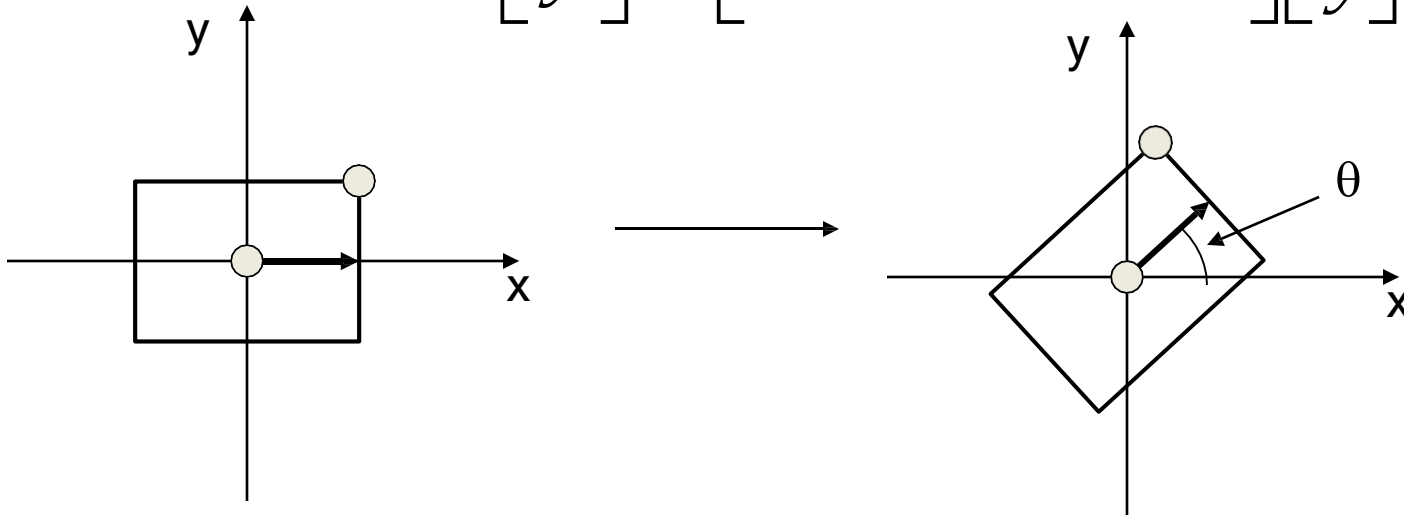
$$\mathbf{R} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$



2D Rotation

- Rotate counter-clockwise about the origin by an angle θ

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$



(b) 2D Rotation:

```
void rotate (point[ ], int n, double degree)
{
    int i;
    double x,y, radians, costheta, sintheta;
    radians = degree * 0.17453293;
    costheta = cos(radians);
    sintheta = sin(radians);
    for (i = 0; i < n; i++)
    {
        x = point[i].x;
        y = point[i].y;
        point[i].x = x*costheta - y * sintheta;
        point[i].y = x* sintheta + y * costheta;
    }
}
```

Example:

Prove that the matrix A is a pure rotation transformation matrix

$$\text{Where, } A = \begin{bmatrix} \frac{1+b}{1-b} & \frac{2}{1-b} \\ \frac{2b}{1-b} & \frac{1+b}{1-b} \end{bmatrix}$$

Solution

$$\text{Det}(A) = \frac{1+b}{1-b} \times \frac{1+b}{1-b} - \frac{2b}{1-b} \times \frac{2}{1-b} = \frac{1+2b+b^2-4b}{(1-b)^2} = \frac{1-2b+b^2}{(1-b)^2} = 1$$

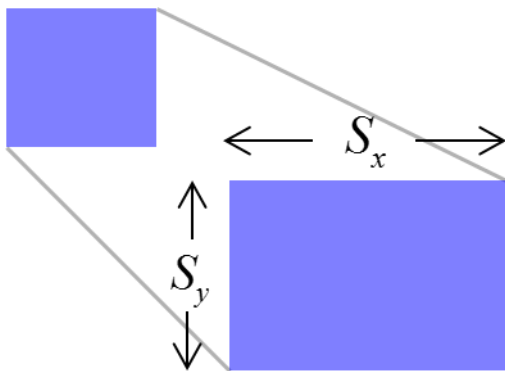
Since $\text{Det}(A)=1$ so the matrix A is a pure rotation

(c) 2D Scaling

- The process of altering the dimensions of an object, *i. e.* enlarging or reducing the size of an object is called scaling.
- The object is scaled by multiplying the coordinates of each endpoint by the scaling factor.
- Let ***s_x***, ***s_y*** be the scale in the positive **x** and **y** directions respectively. Then the scaled vertex is given by

$$x' = x \cdot s_x$$

$$y' = y \cdot s_y$$



$$x' = x \cdot s_x, \quad y' = y \cdot s_y$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} s_x & 0 \\ 0 & s_y \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\mathbf{P}' = \mathbf{S} \cdot \mathbf{P}$$

❖ If ***s_x*** and ***s_y*** values between 0 and 1

- ✓ Point is closer to origin
- ✓ Size decrease

❖ If ***s_x*** and ***s_y*** values are **> 1**

- ✓ Point is away from the origin
- ✓ Size increase

❖ If ***s_x*** and ***s_y*** values are equal

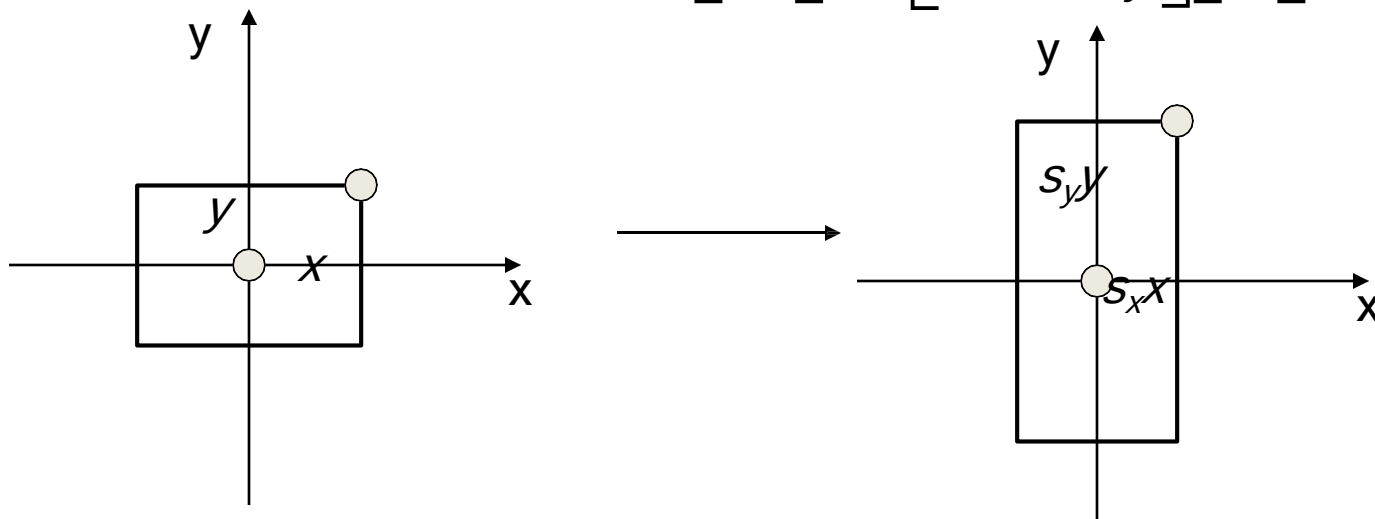
- ✓ Scaling will be uniform



2D Scaling

- Resizes an object in each dimension

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} s_x & 0 \\ 0 & s_y \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$



(c) 2D Scaling:

```
void scale (point[ ], int n, int sx, int sy)
{
    int i;
    for (i = 0; i < n; i++)
    {
        point[i].x = point[i].x * sx;
        point[i].y = point[i].y * sy;
    }
}
```

Example

- Given a square object with coordinate points A(0, 3), B(3, 3), C(3, 0), D(0, 0). Apply the scaling parameter 2 towards X axis and 3 towards Y axis and obtain the new coordinates of the object.

For Coordinates A(0, 3)

$$X_{\text{new}} = X_{\text{old}} \times S_x = 0 \times 2 = 0$$

$$Y_{\text{new}} = Y_{\text{old}} \times S_y = 3 \times 3 = 9$$

For Coordinates B(3, 3)

$$X_{\text{new}} = X_{\text{old}} \times S_x = 3 \times 2 = 6$$

$$Y_{\text{new}} = Y_{\text{old}} \times S_y = 3 \times 3 = 9$$

For Coordinates C(3, 0)

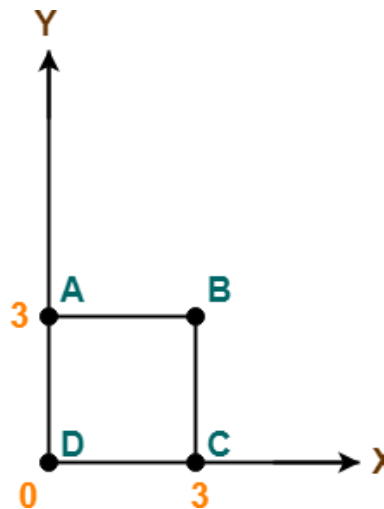
$$X_{\text{new}} = X_{\text{old}} \times S_x = 3 \times 2 = 6$$

$$Y_{\text{new}} = Y_{\text{old}} \times S_y = 0 \times 3 = 0$$

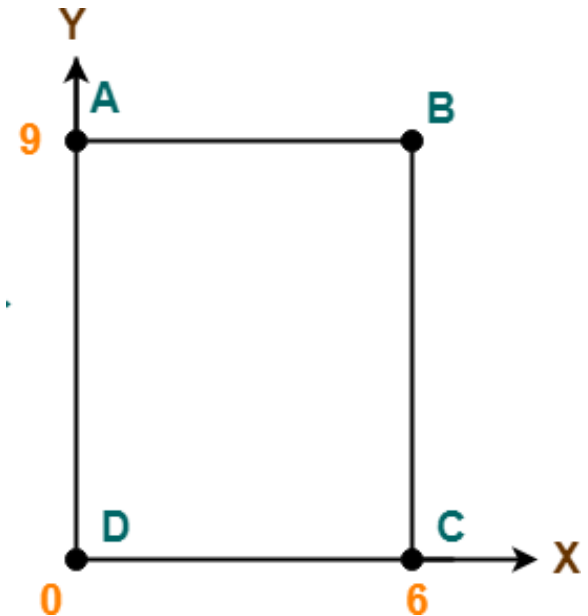
For Coordinates D(0, 0)

$$X_{\text{new}} = X_{\text{old}} \times S_x = 0 \times 2 = 0$$

$$Y_{\text{new}} = Y_{\text{old}} \times S_y = 0 \times 3 = 0$$



Original square



Square after scaling

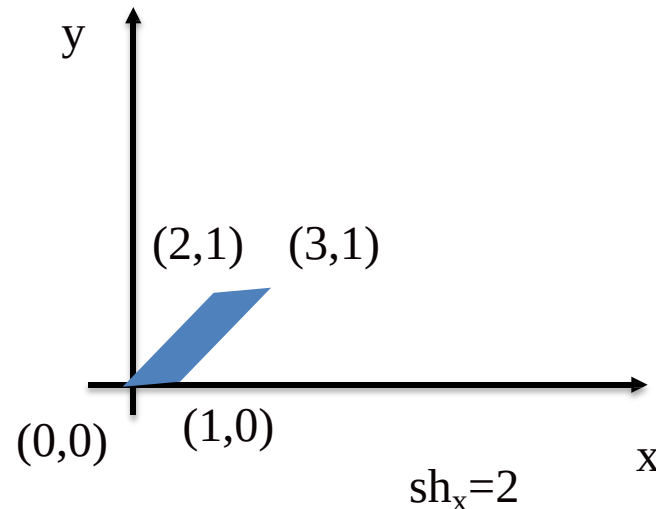
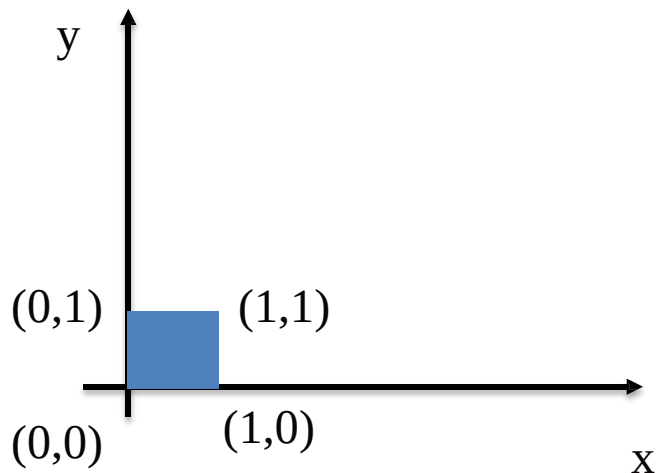


Other 2D Transformations

(Shear)

- **Shear**

- Transformation that distorts the shape of an object such that the transformed shape appears as the object was composed of internal layers that had been caused to slide over each other





Other 2D Transformations

(Shear)

- Shear
 - An x-direction shear relative to the x axis

$$\begin{bmatrix} 1 & sh_x \\ 0 & 1 \end{bmatrix}$$

$$x' = x + sh_x \cdot y$$

$$y' = y$$

- An y-direction shear relative to the y axis

$$\begin{bmatrix} 1 & 0 \\ sh_y & 1 \end{bmatrix}$$

$$x' = x$$

$$y' = y + sh_y \cdot x$$

$$x$$

- Shear relative to the origin

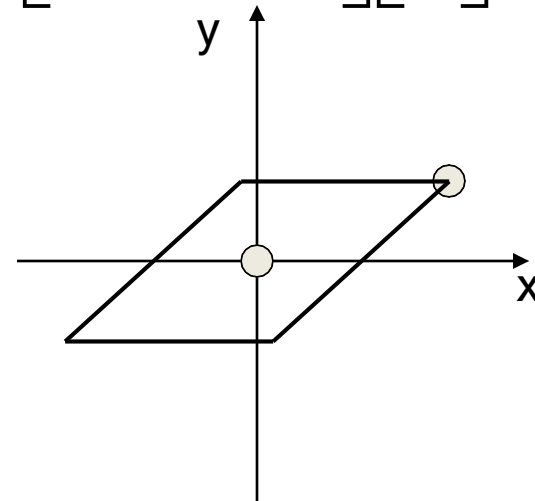
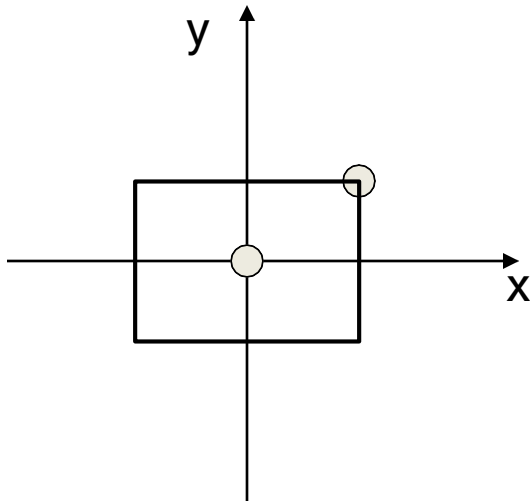
$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & sh_x \\ sh_y & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$



X-Axis Shear

- Shear along x axis (What is the matrix for y axis shear?)

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & sh_x \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$



Example

- Given a triangle with points (1, 1), (0, 0) and (1, 0). Apply shear parameter 2 on X axis and 2 on Y axis and find out the new coordinates of the object.

Shearing in X Axis-

For Coordinates A(1, 1)

$$X_{\text{new}} = X_{\text{old}} + \text{Shx} \times Y_{\text{old}} = 1 + 2 \times 1 = 3$$

$$Y_{\text{new}} = Y_{\text{old}} = 1$$

Thus, New coordinates of point A after shearing = (3, 1).

For Coordinates B(0, 0)

$$X_{\text{new}} = X_{\text{old}} + \text{Shx} \times Y_{\text{old}} = 0 + 2 \times 0 = 0$$

$$Y_{\text{new}} = Y_{\text{old}} = 0$$

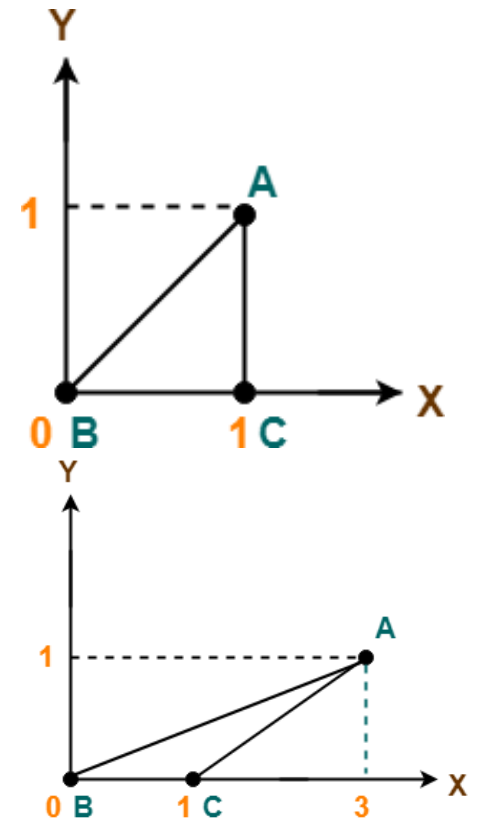
Thus, New coordinates of corner B after shearing = (0, 0).

For Coordinates C(1, 0)

$$X_{\text{new}} = X_{\text{old}} + \text{Shx} \times Y_{\text{old}} = 1 + 2 \times 0 = 1$$

$$Y_{\text{new}} = Y_{\text{old}} = 0$$

Thus, New coordinates of corner C after shearing = (1, 0).



Shearing in X Axis

Example

- Given a triangle with points (1, 1), (0, 0) and (1, 0). Apply shear parameter 2 on X axis and 2 on Y axis and find out the new coordinates of the object.

Shearing in Y Axis-

For Coordinates A(1, 1)

$$X_{\text{new}} = X_{\text{old}} = 1$$

$$Y_{\text{new}} = Y_{\text{old}} + Sh_y \times X_{\text{old}} = 1 + 2 \times 1 = 3$$

Thus, New coordinates of corner A after shearing = (1, 3).

For Coordinates B(0, 0)

$$X_{\text{new}} = X_{\text{old}} = 0$$

$$Y_{\text{new}} = Y_{\text{old}} + Sh_y \times X_{\text{old}} = 0 + 2 \times 0 = 0$$

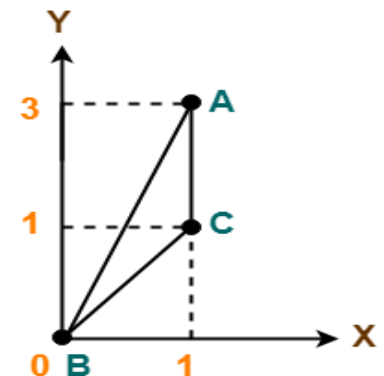
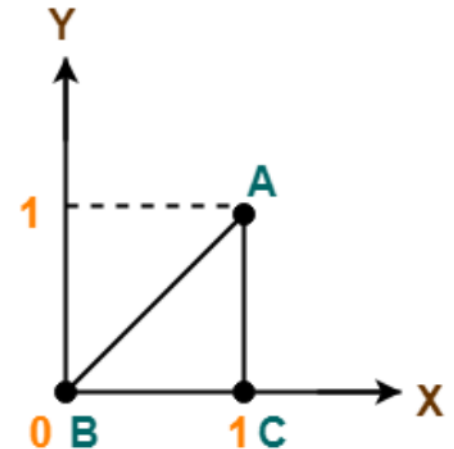
Thus, New coordinates of corner B after shearing = (0, 0).

For Coordinates C(1, 0)

$$X_{\text{new}} = X_{\text{old}} = 1$$

$$Y_{\text{new}} = Y_{\text{old}} + Sh_y \times X_{\text{old}} = 0 + 2 \times 1 = 2$$

Thus, New coordinates of corner C after shearing = (1, 2).



Shearing in Y Axis



2D Shear

Example

Show that the shear transformation in x and y directions together is not the same as shear along x followed by shear along y?

Ans:

$$\begin{bmatrix} 1 & b \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ c & 1 \end{bmatrix} = \begin{bmatrix} 1+bc & b \\ c & 1 \end{bmatrix} \neq \begin{bmatrix} 1 & b \\ c & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 \\ c & 1 \end{bmatrix} \begin{bmatrix} 1 & b \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & b \\ c & 1+bc \end{bmatrix} \neq \begin{bmatrix} 1 & b \\ c & 1 \end{bmatrix}$$

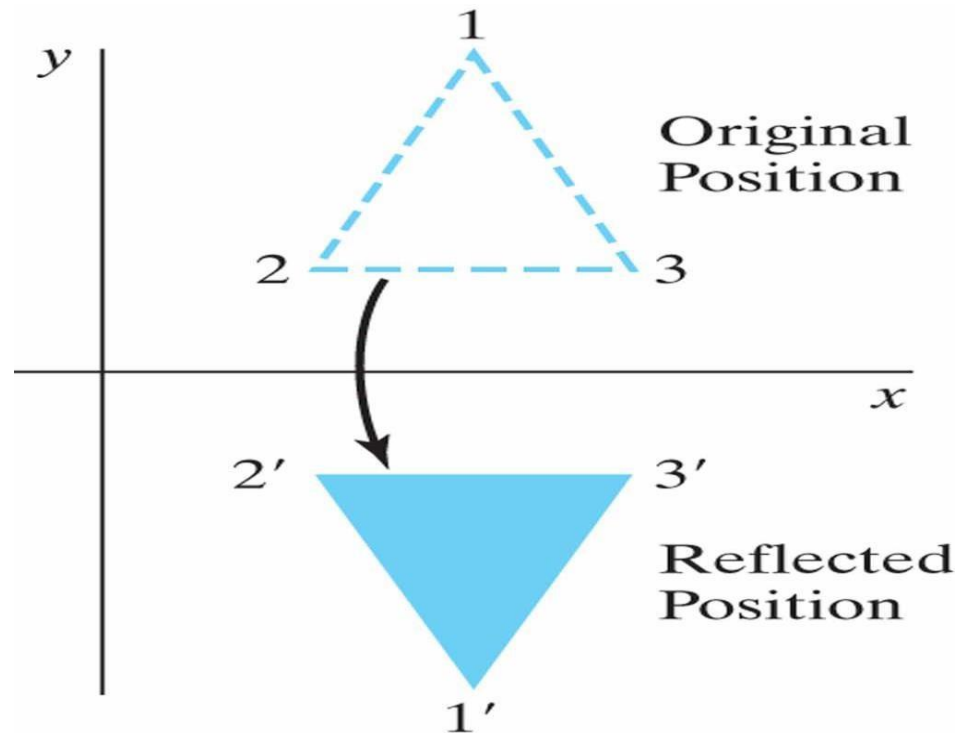


Other 2D Transformations

(Reflection)

- **Reflection**

- Transformation that produces a mirror image of an object

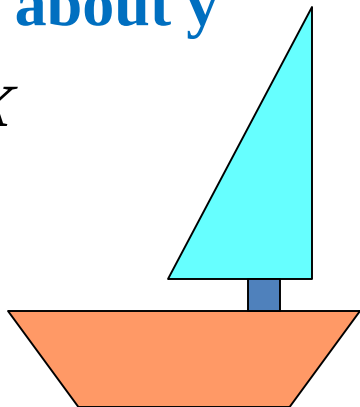


2D Reflections

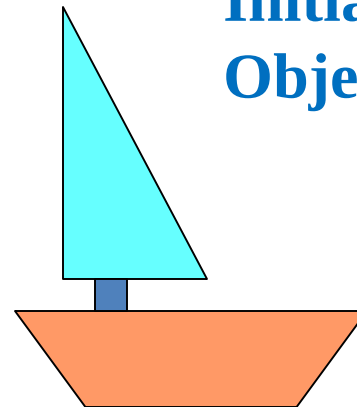
Reflection about y

$$X' = -X$$

$$y' = y$$



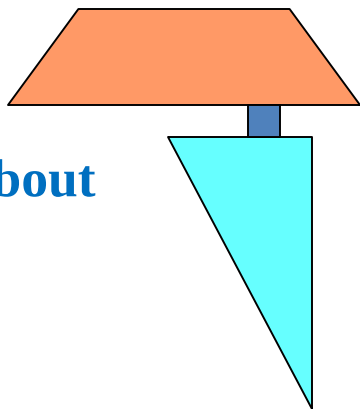
Initial
Object



Reflection about
origin

$$X' = -X$$

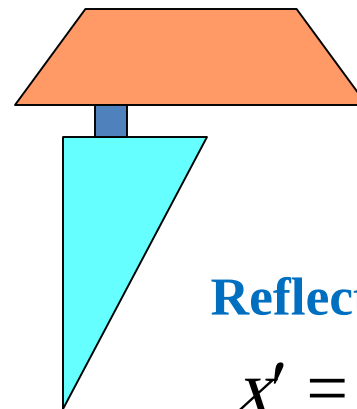
$$y' = -y$$



Reflection about x

$$X' = X$$

$$y' = -y$$



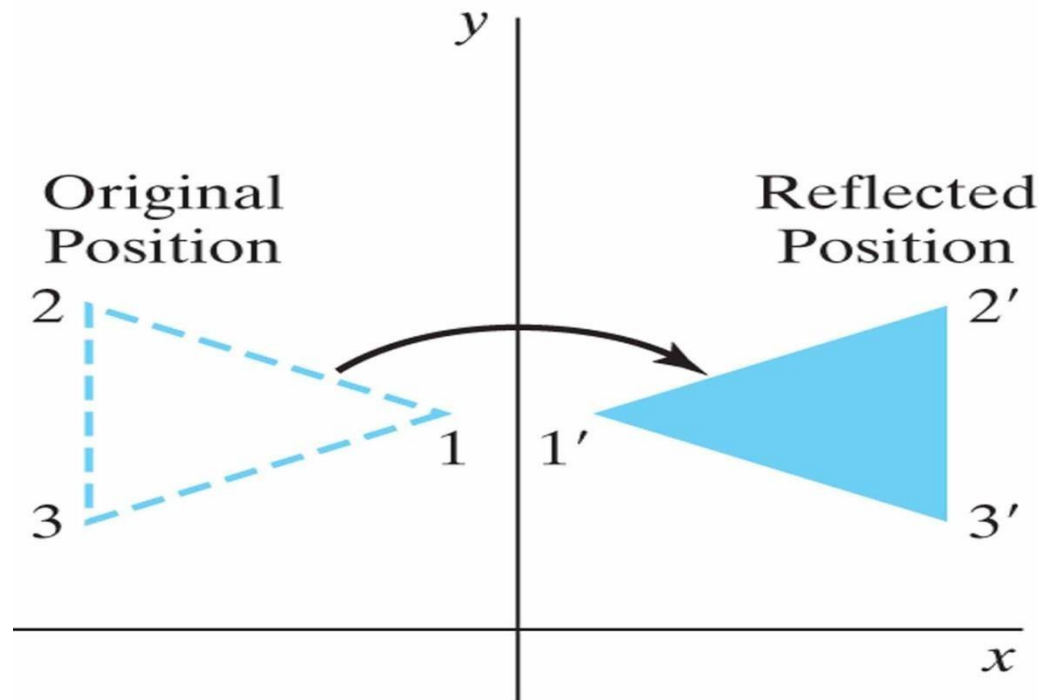


Other 2D Transformations

(Reflection)

- Reflection
 - Reflection about the line $x=0$ (the y axis)

$$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$





Other 2D Transformations

(Reflection)

- Reflection
 - Image is generated relative to an axis of reflection by rotating the object 180° about the reflection axis
 - **Reflection about the line $y=0$ (the x axis) (previous slide)**

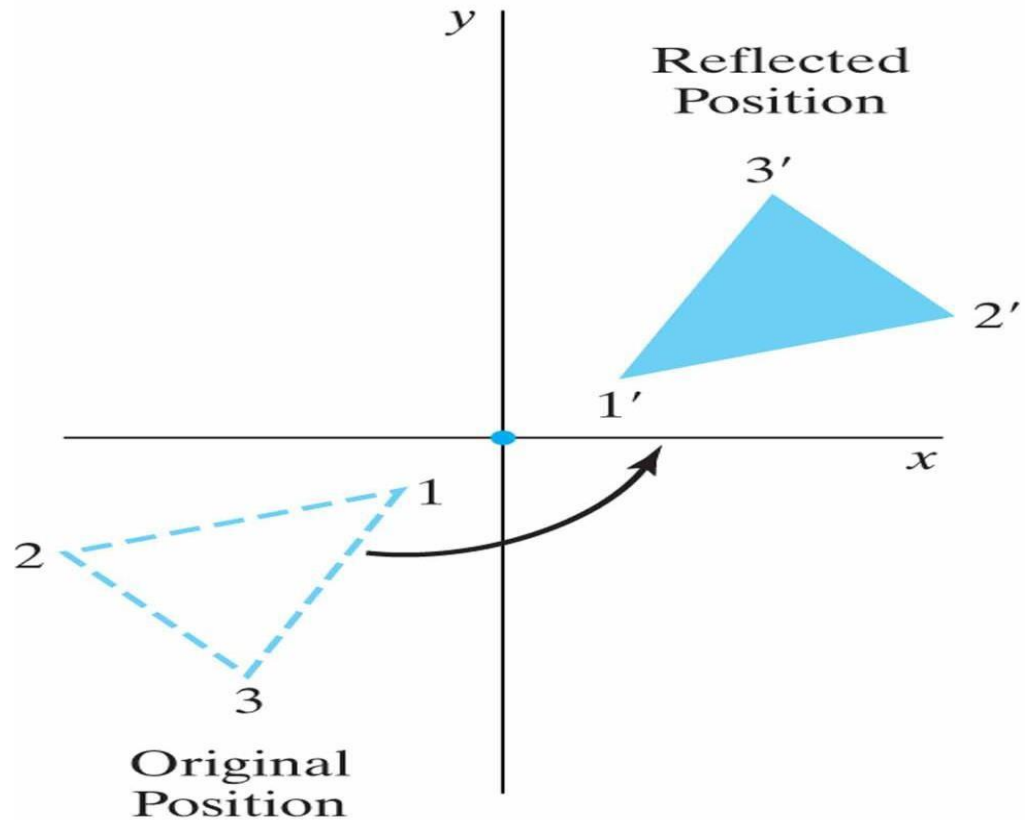
$$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Other 2D Transformations

(Reflection)

- Reflection about the origin

$$\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$



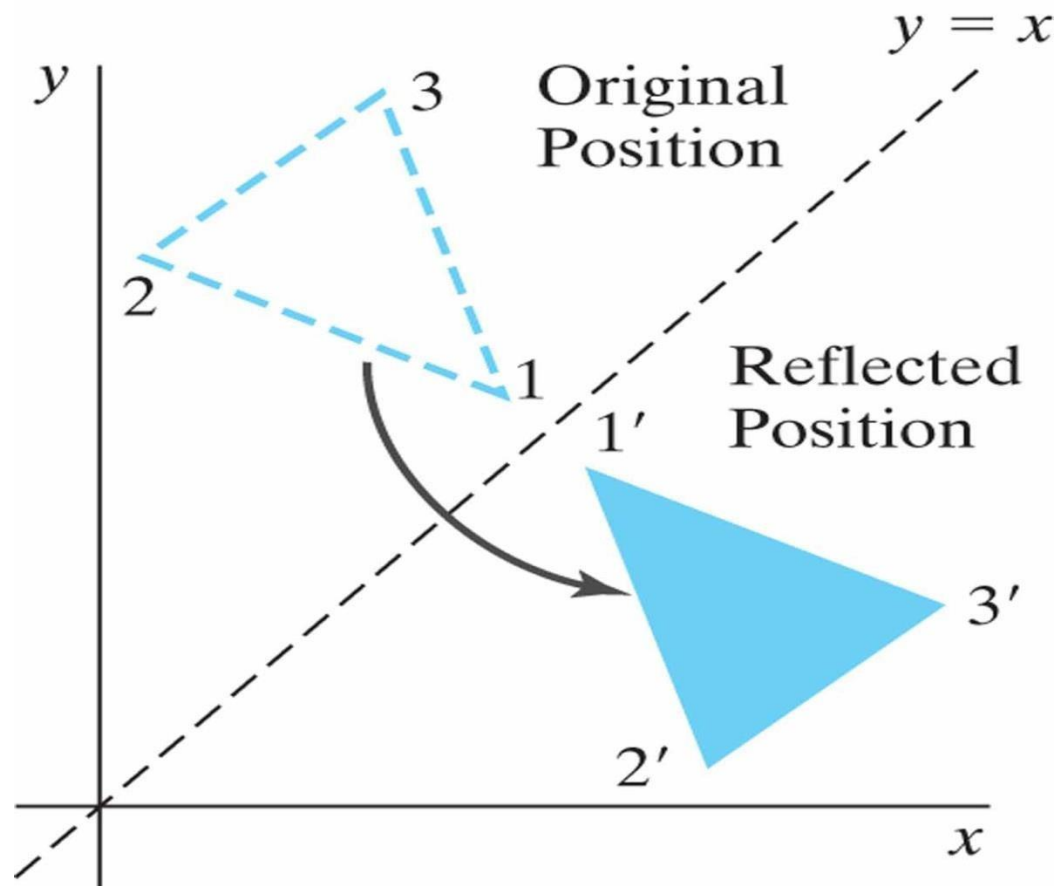
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Other 2D Transformations

(Reflection)

- Reflection about the line $y=x$

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

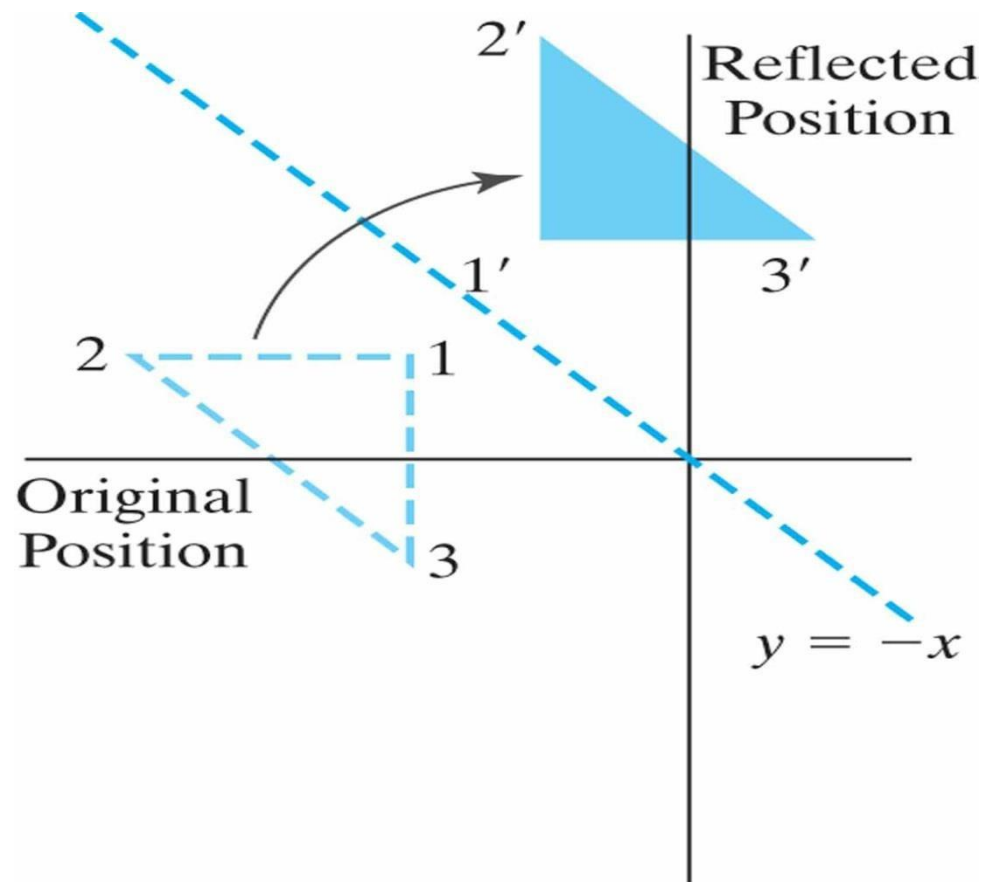


Other 2D Transformations

(Reflection)

- Reflection about the line $y=-x$

$$\begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$$





Other 2D Transformations

(Reflection Example)

- Given a triangle with coordinate points A(3, 4), B(6, 4), C(5, 6). Apply the reflection on the X axis and obtain the new coordinates of the object.

Reflection on X Axis-

For Coordinates A(3, 4)

$$X_{\text{new}} = X_{\text{old}} = 3$$

$$Y_{\text{new}} = -Y_{\text{old}} = -4$$

For Coordinates B(6, 4)

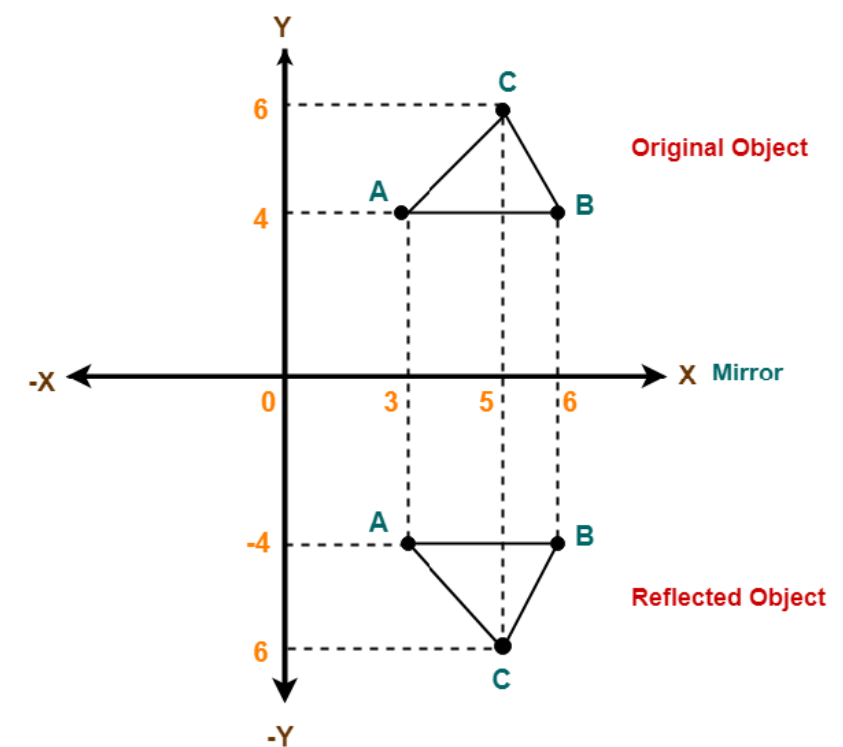
$$X_{\text{new}} = X_{\text{old}} = 6$$

$$Y_{\text{new}} = -Y_{\text{old}} = -4$$

For Coordinates C(5, 6)

$$X_{\text{new}} = X_{\text{old}} = 5$$

$$Y_{\text{new}} = -Y_{\text{old}} = -6$$



Example:

A unit square is transformed by 2 x 2 transformation matrix. The resulting position vector are :-

$$\begin{pmatrix} 0 & 2 & 8 & 6 \\ 0 & 3 & 4 & 1 \end{pmatrix}, \text{ what is the transformation matrix?}$$

Ans: Suppose the unit square have coordinates

$$(x, y)$$

$$(x+1, y)$$

$$(x+1, y+1)$$

$$(x, y+1)$$

and let the transformation matrix be $\begin{pmatrix} a & c \\ b & d \end{pmatrix}$

$$\text{So, } \begin{pmatrix} 0 & 2 & 8 & 6 \\ 0 & 3 & 4 & 1 \end{pmatrix} = \begin{pmatrix} a & c \\ b & d \end{pmatrix} \begin{pmatrix} x & x+1 & x+1 & x \\ y & y & y+1 & y+1 \end{pmatrix}$$

$$= \begin{pmatrix} ax+cy & a(x+1)+cy & a(x+1)+c(y+1) & ax+c(y+1) \\ bx+by & b(x+1)+dy & b(x+1)+d(y+1) & bx+d(y+1) \end{pmatrix}$$

Example:

$$\text{Now, } ax+cy=0 \quad \text{and} \quad bx+cy=0$$

$$a(x+1)+cy=2 \quad \text{and} \quad b(x+1)+dy=3$$

$$a(x+1)+c(y+1)=8 \quad \text{and} \quad b(x+1)+d(y+1)=4$$

$$ax+c(y+1)=6 \quad \text{and} \quad bx+d(y+1)=1$$

from this we get,

$$a=2, \quad b=3, \quad c=6, \quad d=1$$

Thus, the transformation matrix is $\begin{pmatrix} 2 & 6 \\ 3 & 1 \end{pmatrix}$

Translation (OpenGL)

Specifying a 2D-Translation:

```
glTranslatef(tx, ty, 0.0);
```

(The z component is set to 0 for 2D translation).

Rotation About the Origin

Specifying a 2D-Rotation about the origin:

```
glRotatef(theta, 0.0, 0.0, 1.0);
```

theta: Angle of rotation in degrees.

The above function defines a rotation about the z-axis (0,0,1).

Scaling About the Origin

Specifying a 2D-Scaling with respect to the origin:

```
glScalef(sx, sy, 1.0);
```

s_x , s_y : Scale factors along x, y.

For proper scaling s_x , s_y must be positive.

For 2D scaling, the third scale factor must be set to 1.0.

Reflection